

Prepared by: Abiola Aderemi | April 2026
Tools: Python (scikit-learn) & Microsoft Power BI.
Dataset: UCI Machine Learning Repository.

PREDICTIVE INTELLIGENCE IN BANK MARKETING

Leveraging a Decision Tree Classifier to Transform Campaign Targeting and Maximize Term Deposit Conversions.

DATASET

45,211

Customer Records

**MODEL
ACCURACY**

89.9%

**DISCRIMINATIVE
POWER**

0.90

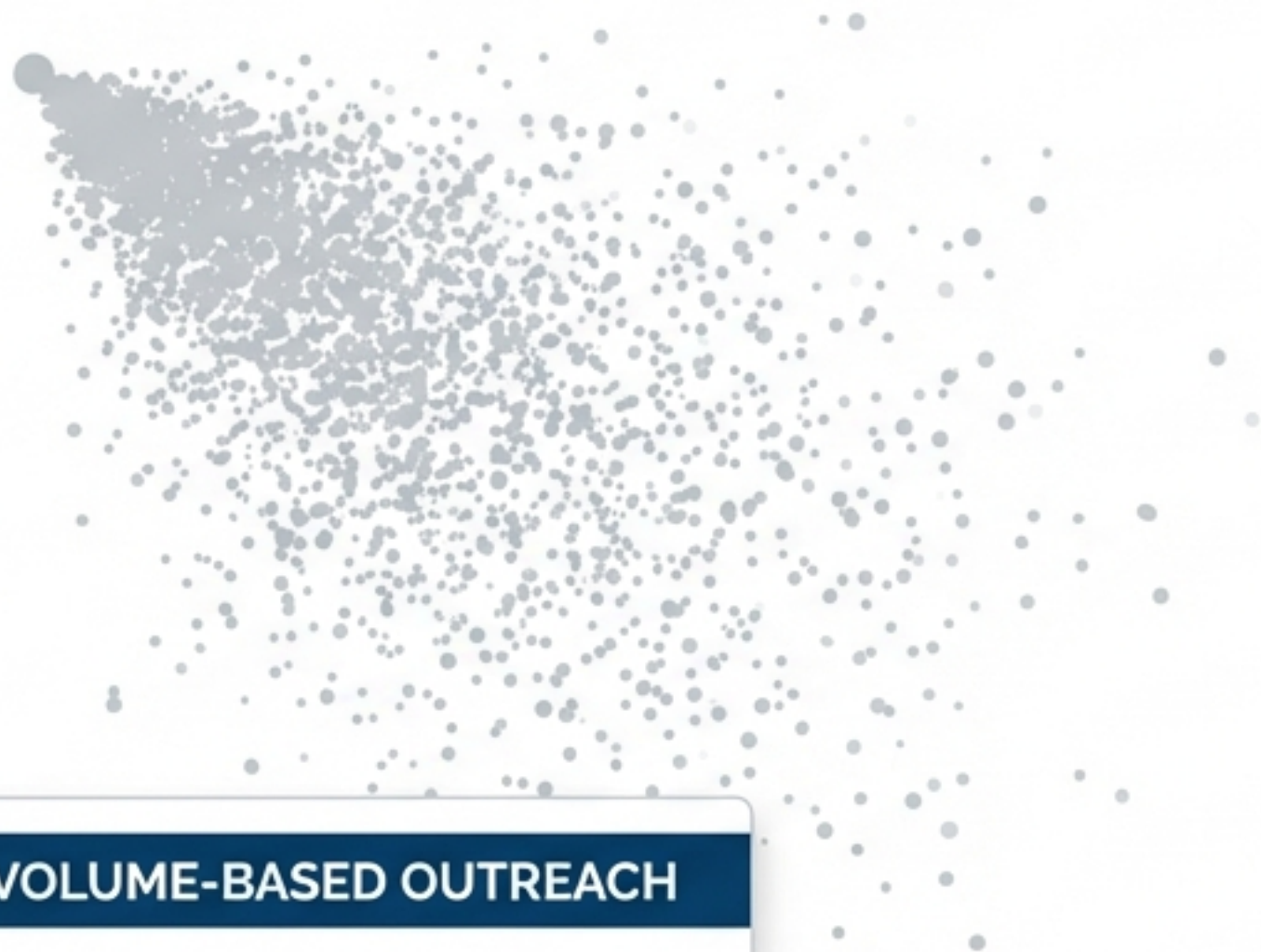
ROC-AUC

**COMMERCIAL
IMPACT**

5X

Efficiency Gain

SPRAY-AND-PRAY MARKETING IS A DRAIN ON RESOURCES



VOLUME-BASED OUTREACH

Wastes agent time, burns marketing budget, and irritates non-receptive customers.

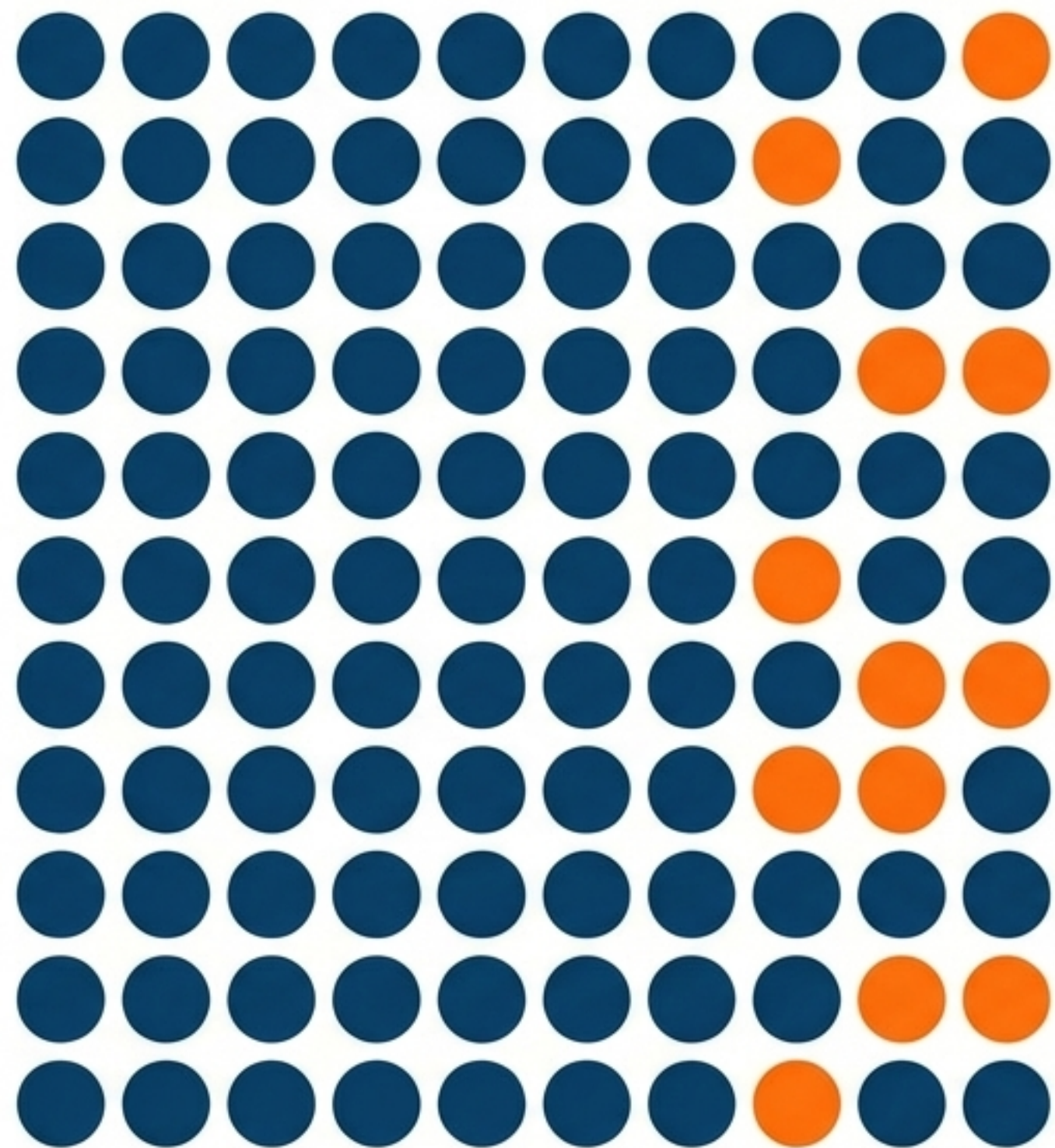


PRECISION TARGETING

Predict subscription probability before a single call is made.

THE OBJECTIVE: Term deposits are stable, multi-year funding sources. Every successful conversion is a multi-year financial relationship. We must shift from reactive volume to proactive precision.

THE NEEDLE IN A HAYSTACK DATA CHALLENGE



88.3%
Non-Subscribers

39,922 records

11.7%
Subscribers

5,289 records

THE IMBALANCE REALITY

A naive model that predicts "No" for everyone would achieve 88.3% accuracy without learning a single useful pattern.

THE SOLUTION

Stratified 80/20 train-test splitting to prevent bias, and utilizing ROC-AUC to measure genuine discriminative power rather than superficial accuracy.

THE END-TO-END ANALYTICS PIPELINE



1. PREPROCESSING

Categorical encoding of 9 columns. Zero missing values required, no imputation.



2. STRATIFICATION

80/20 train-test split (36,168 train / 9,043 test) strictly maintaining the 88.3/11.7 class ratio.



3. MODELING

Decision Tree Classifier selected, optimized via GridSearchCV.



4. BUSINESS INTELLIGENCE

Model predictions and features exported to interactive dashboards for front-line campaign managers.

WHY A DECISION TREE CLASSIFIER?



TUNING THE MODEL FOR REAL-WORLD REALITY

BASELINE (UNPRUNED)

 **DEPTH: 29**

 **LEAVES: 2,985**

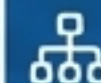
 **ROC-AUC: 0.704**

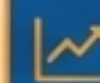


Overfitted: Memorized the training data; failed to generalize.

OPTIMIZED (GRIDSEARCHCV)

 **DEPTH: 10**

 **LEAVES: 181**

 **ROC-AUC: 0.901**



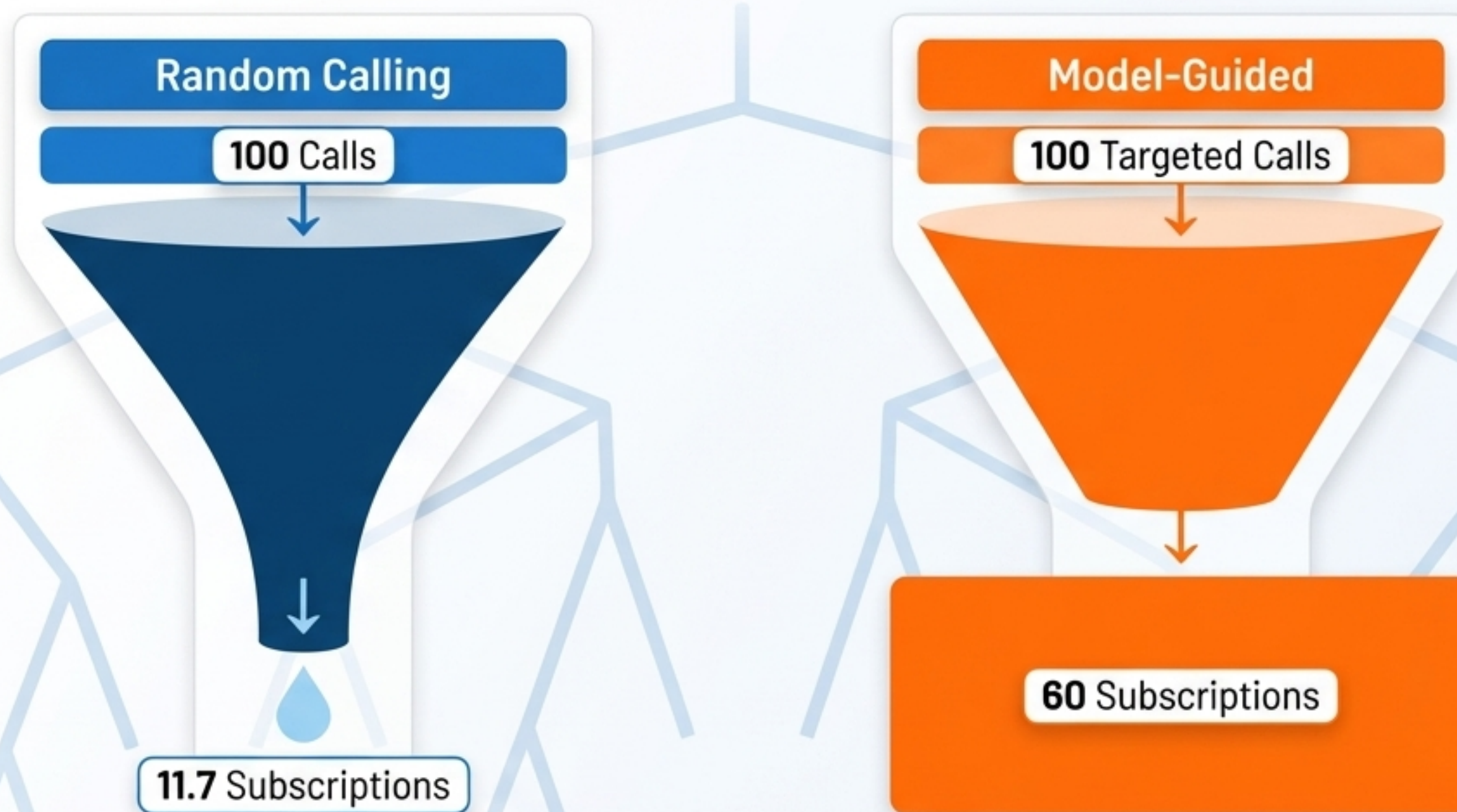
Generalizable: Controlled complexity; highly predictive on unseen data.

Insight: Hyperparameter tuning gives the model boundaries, ensuring it learns underlying human behavioral patterns rather than memorizing specific past customers.

THE 5x COMMERCIAL EFFICIENCY MULTIPLIER

89.9%
Test Accuracy

0.90
ROC-AUC



Precision is **60%**, Recall is **40%**.

The **Commercial Reality**: If we restrict agent outreach exclusively to customers the model flags, **60 out of 100** will convert. This reduces required call volumes by **5x** to achieve the identical number of total subscriptions.

THE ANATOMY OF A CONVERTER: WHAT DRIVES SUBSCRIPTION?



1. Duration (0.481)

The dominant signal. Accounts for **48%** of predictive power.

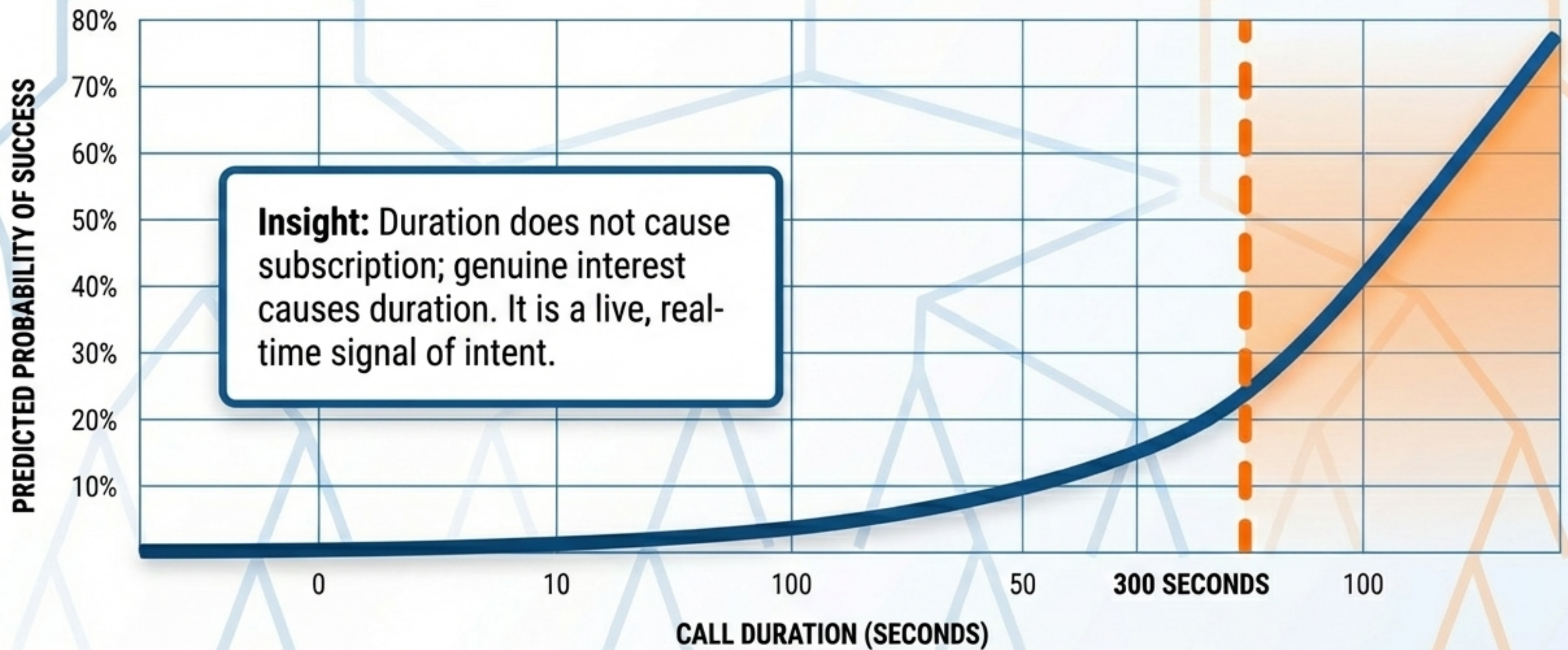
2. Month (0.143): Severe seasonal variance in receptiveness.

3. Contact Type (0.115): Cellular dominance over landlines.

4. Recency (0.082): Days since last campaign contact.

5. Housing Loan (0.056): Lack of a housing loan indicates disposable capital.

THE GOLDEN SIGNAL: THE 300-SECOND ACTIVATION THRESHOLD



OPERATIONAL TACTIC: THE 5-MINUTE MARK IS CRITICAL. AGENTS MUST BE TRAINED TO RECOGNIZE WHEN A CALL CROSSES THIS THRESHOLD AND PIVOT THEIR SCRIPT FROM 'PITCHING' THE PRODUCT TO 'CLOSING' THE SALE.

SEASONAL MISALIGNMENT IN CAMPAIGN TIMING

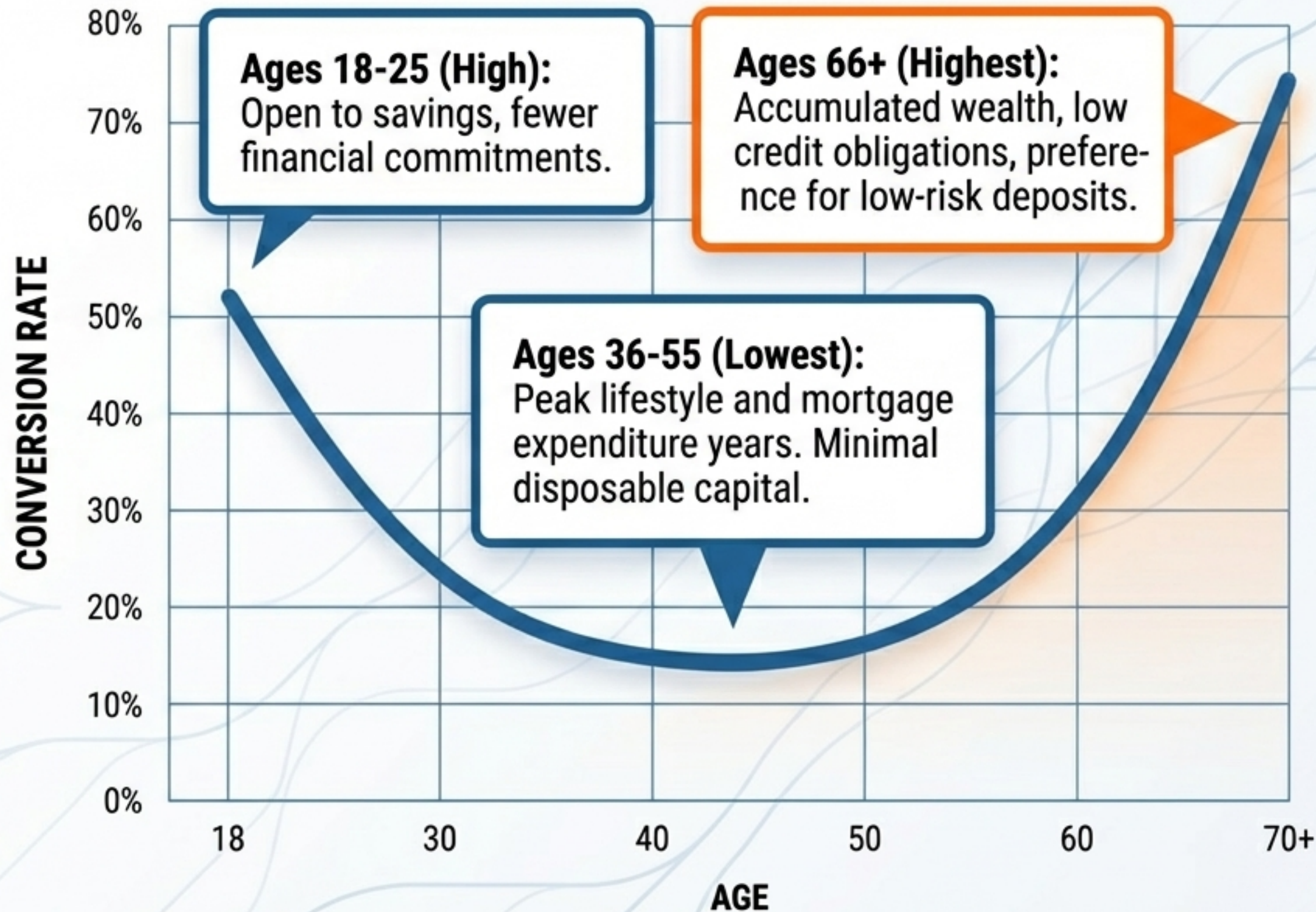


Insight: **Spring and Summer** months yield the lowest returns despite high historical campaign effort.



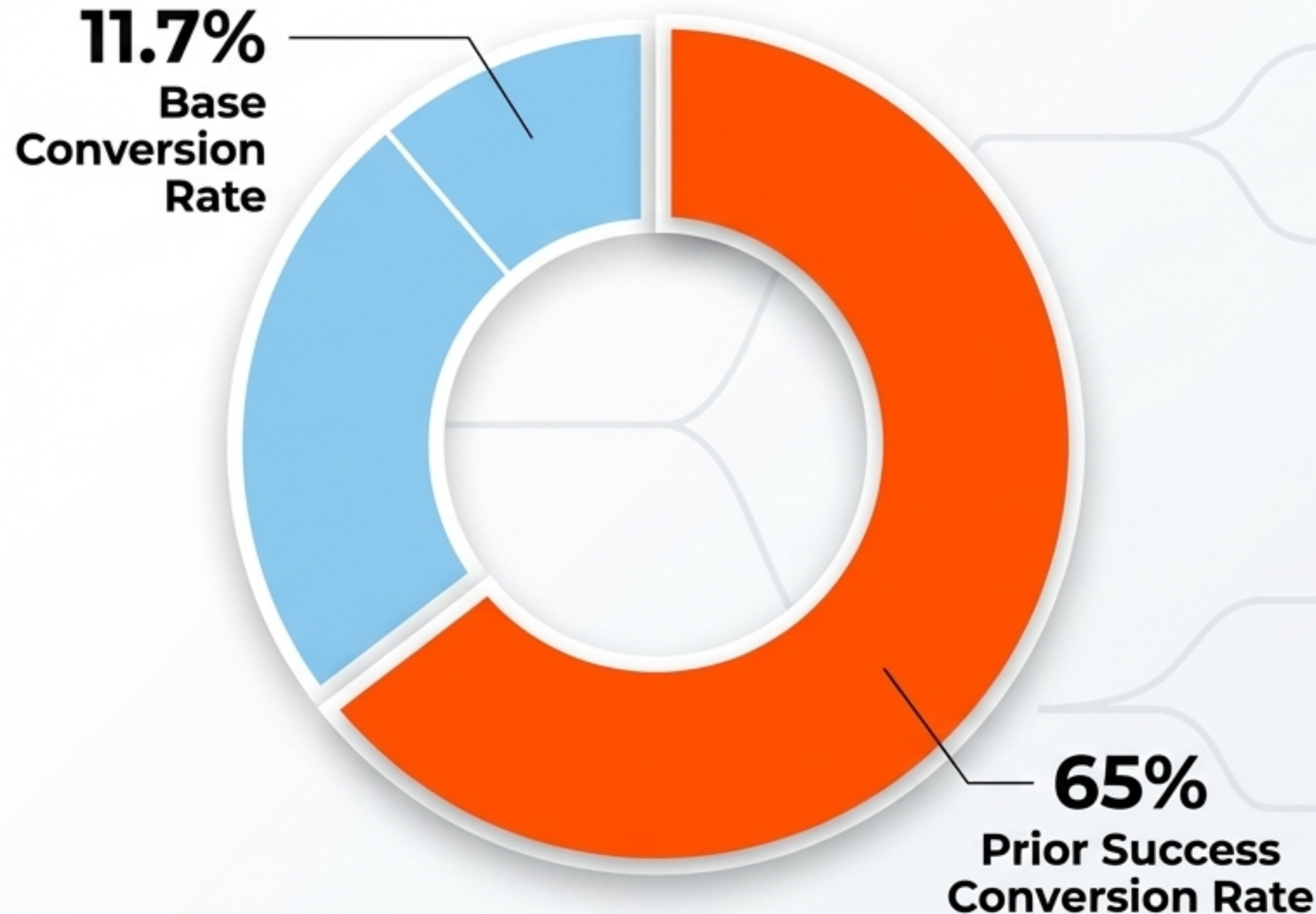
Action Directive: Reallocate marketing budget and agent capacity away from the summer slump, concentrating resources heavily on **Autumn, Winter, and the end-of-Q1.**

THE LIFE-STAGE ECONOMICS OF A SUBSCRIBER



Occupation
TOP PERFORMERS
Retirees, Students, Management
BOTTOM PERFORMERS
Blue-collar, Services

THE 65% RE-ENGAGEMENT MULTIPLIER



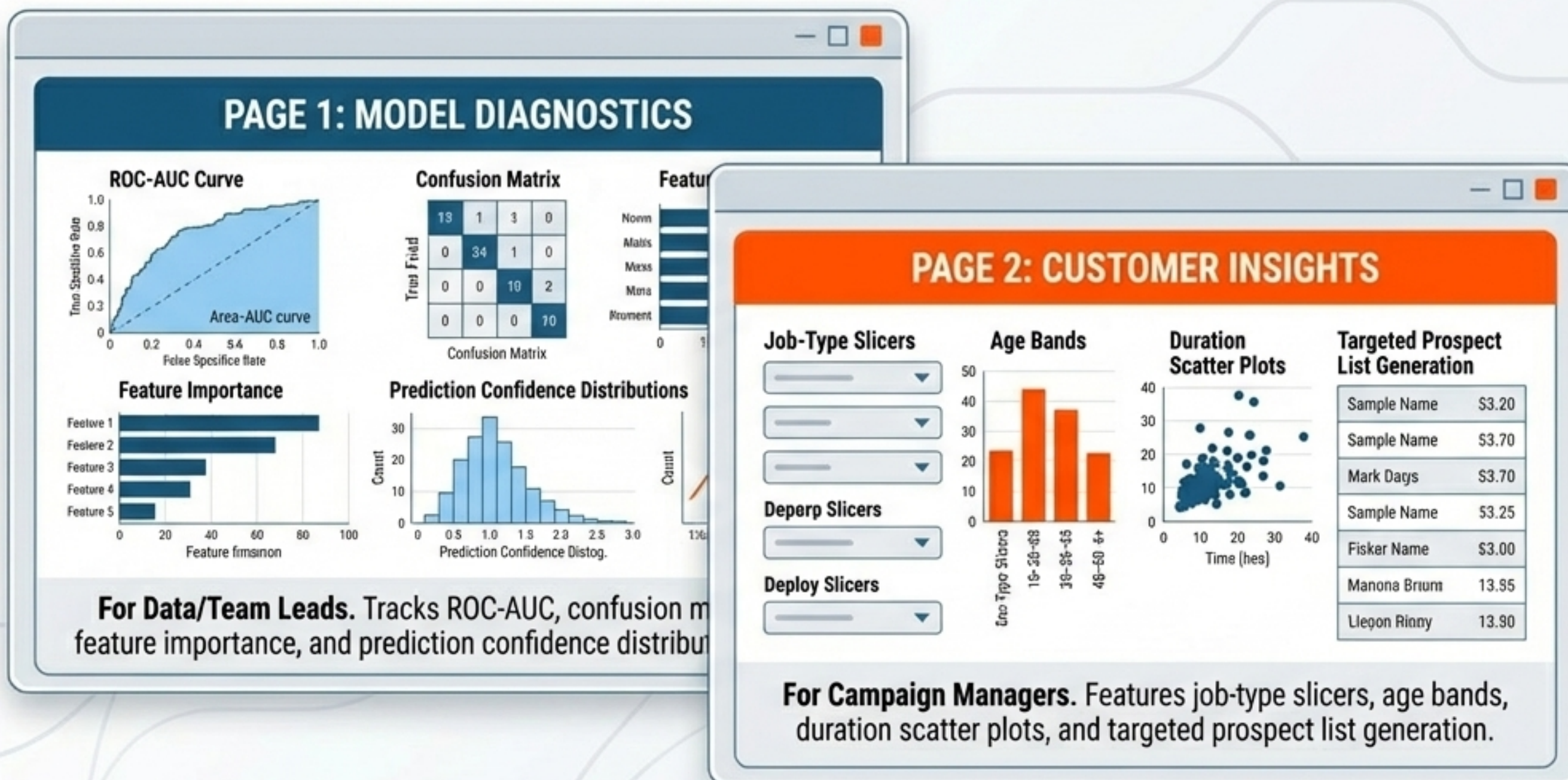
THE INSIGHT:

Customers who subscribed in a previous campaign convert at an astonishing 65% rate in new campaigns—nearly 6 times the baseline.

STRATEGIC IMPERATIVE:

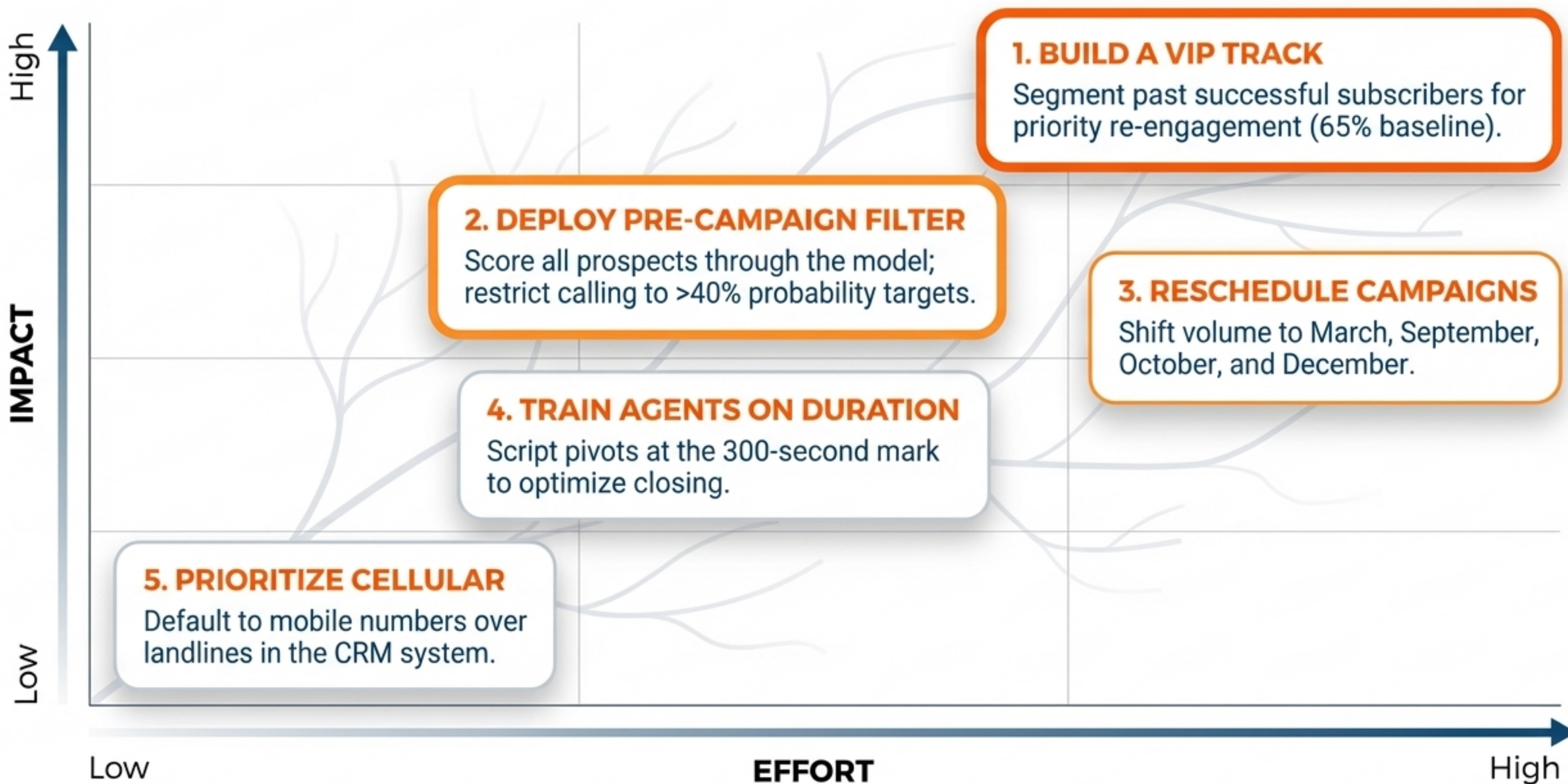
A customer who said 'yes' before is the most likely to say 'yes' again. Never treat them as cold leads. Prioritize a dedicated VIP re-engagement track.

DELIVERING INTELLIGENCE TO THE FRONT LINE



IMPACT: Breaking the data science silo. Insights are extracted via Python and placed directly into operational hands via interactive dashboards.

FIVE STRATEGIC ACTIONS FOR IMMEDIATE ROI



A SCALABLE FRAMEWORK FOR PREDICTIVE STRATEGY

THE ACHIEVEMENT

Replaced intuition and spray-and-pray marketing with a reproducible, highly accurate decision framework (89.9% Accuracy).



THE ASSET

The end-to-end Python-to-PowerBI pipeline (cleaning, engineering, modeling, BI) is fully scalable and owned by the bank.



NEXT FRONTIERS

This exact methodology is ready for immediate deployment in: Fintech Credit Scoring, Telecoms Churn Prediction, Lifetime Value Optimization.